

Erdős Problem #935

EP-935

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For any integer $n = \prod p^{k_p}$ let $Q_2(n)$ be the powerful part of n , so that $Q_2(n) = \prod_{k_p \geq 2} p^{k_p}$. Is it true that, for every $\epsilon > 0$ and $\ell \geq 1$, if n is sufficiently large then $Q_2(n(n+1) \cdots (n+\ell)) < n^{2+\epsilon}$? If $\ell \geq 2$ then is $\limsup_{n \rightarrow \infty} \frac{Q_2(n(n+1) \cdots (n+\ell))}{n^2}$ infinite?

If $\ell \geq 2$ then is $\lim_{n \rightarrow \infty} \frac{Q_2(n(n+1) \cdots (n+\ell))}{n^{\ell+1}} = 0$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>